**In this question you must show all stages of your working.**
**Solutions relying on calculator technology are not acceptable.**

Given that

- the point $A$ has coordinates $\left(-2\sqrt{3}, 5\right)$
- the point $B$ has coordinates $\left(7\sqrt{3}, 8\right)$
- the straight line $l_1$ passes through $A$ and $B$

(a) show that the gradient of $l_1$ is $p\sqrt{3}$, where $p$ is a rational constant to be found.
You must show each step of your working.

**(2)**

The straight line $l_2$ is perpendicular to $l_1$ and passes through $A$.

(b) Find the equation of $l_2$, giving your answer in the form $y = mx + c$, where $m$ and $c$ are constants.

**(3)**